

Exchange-Coulomb block-encoding resource sweep

Parameters: $N_{\text{up}} = 4$, $N_{\text{down}} = 22$, $N_{\text{IP}} = 208$
 $N_k = k^3$ for k in $[1, 2, 3, 4, 5, 6]$ (so N_k in $[1, 8, 27, 64, 125, 216]$)
phase_bitsize $b = 32$

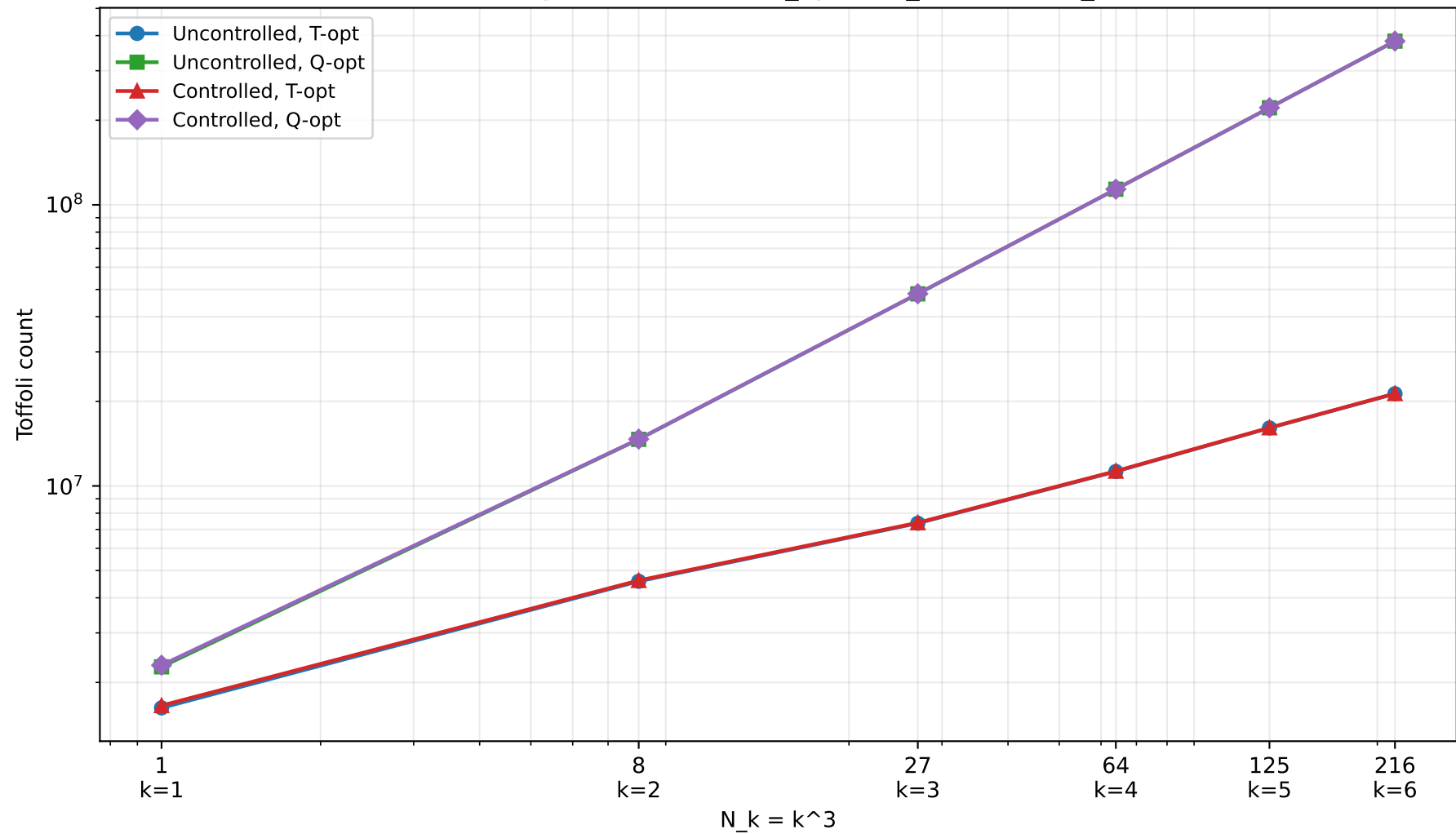
Construction (data-free):

$B = (B_{\text{up}} \otimes B_{\text{down}}) \cdot \text{ModSub} \cdot \text{PrepareUniformSuperposition}_{\text{adj}}$
Each B_{up} , B_{down} is a `ReflectionRectangularBlockEncoding`
 $\text{Full} = B \cdot C \cdot B^\dagger$ where C is another `ReflectionRectangularBlockEncoding`
on the inner matrix register ($\sum_Q |Q\rangle\langle Q| \otimes A'_Q$).

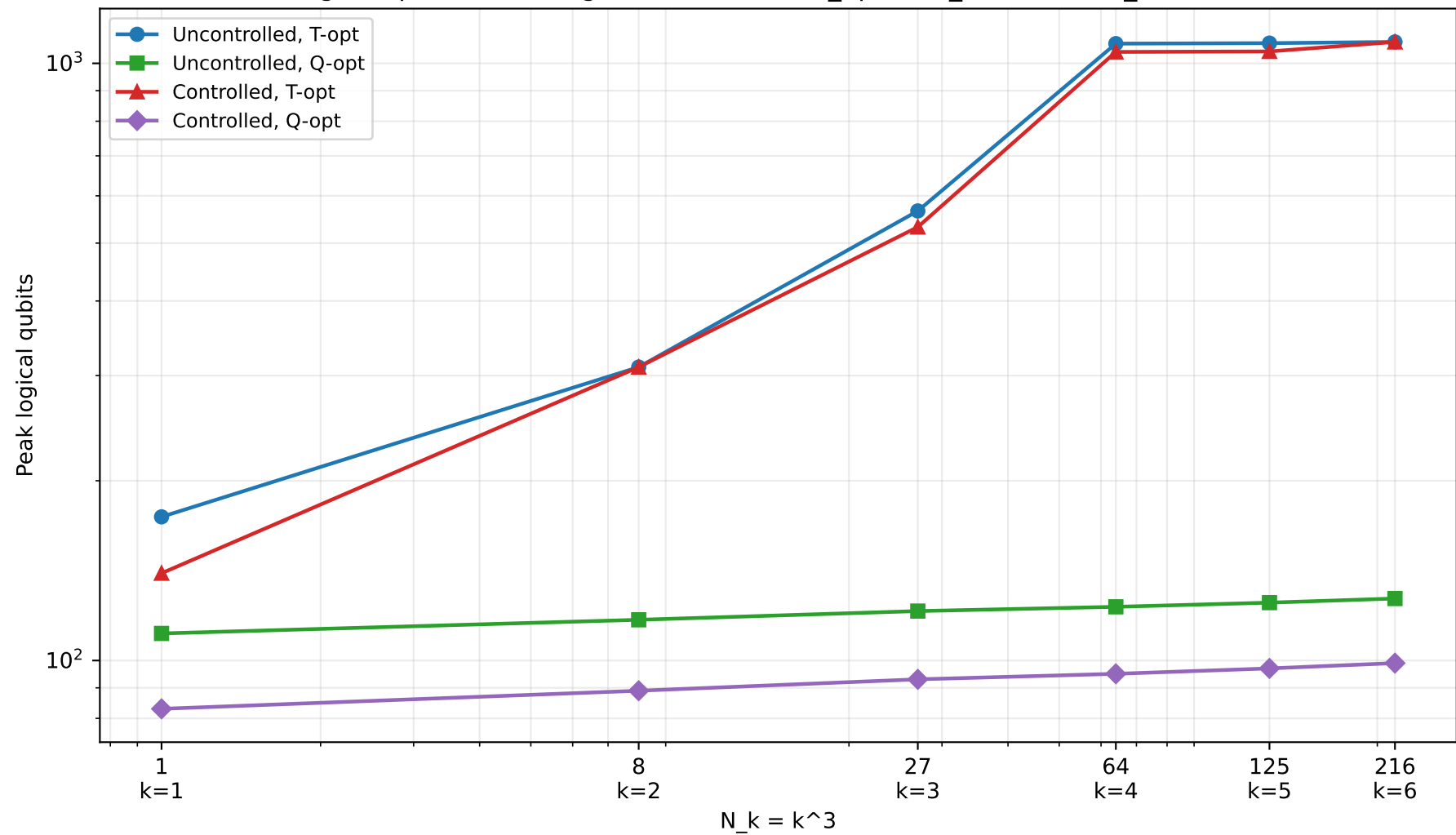
Controlled version exploits the B-sandwich: only C is promoted to a controlled block encoding; outer $B / B^\dagger$ pair self-cancels at $\text{ext_ctrl} = 0$.

All Toffoli and qubit counts come from Qualtran's `QECGatesCost / QubitCount` walking the bloq's `build_call_graph` (no analytic formulas).

Toffoli count: exchange-Coulomb BE ($N_{\text{up}}=4$, $N_{\text{down}}=22$, $N_{\text{IP}}=208$, $b=32$)



Peak logical qubits: exchange-Coulomb BE ($N_{\text{up}}=4$, $N_{\text{down}}=22$, $N_{\text{IP}}=208$, $b=32$)



Resource counts: exchange-Coulomb BE

k	N_k	Uncontrolled T _{top}	Controlled T _{top}	Uncontrolled Q _{top}	Controlled Q _{top}	Uncontrolled T _{opt}	Controlled T _{opt}	Uncontrolled Q _{opt}	Controlled Q _{opt}
1	1	1,623,654	174	2,271,913	111	1,654,464	140	2,302,723	83
2	8	4,579,178	310	14,645,677	117	4,609,988	310	14,676,487	89
3	27	7,366,670	566	48,231,629	121	7,397,480	532	48,262,439	93
4	64	11,257,168	1079	113,635,763	123	11,287,978	1045	113,666,573	95
5	125	16,078,040	1081	221,464,285	125	16,108,850	1047	221,495,095	97
6	216	21,282,624	1086	382,323,155	127	21,313,434	1086	382,353,965	99